

2.1

(a) $T(x[n]) = g[n]x[n]$

- Stable: Let $|x[n]| \leq M$ then $|T(x[n])| \leq |g[n]|M$. So, it is stable if $|g[n]|$ is bounded.
- Causal: $y_1[n] = g[n]x_1[n]$ and $y_2[n] = g[n]x_2[n]$, so if $x_1[n] = x_2[n]$ for all $n < n_0$, then $y_1[n] = y_2[n]$ for all $n < n_0$, and the system is causal.
- Linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= g[n](ax_1[n] + bx_2[n]) \\ &= ag[n]x_1[n] + bg[n]x_2[n] \\ &= aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

So this is linear.

- Not time-invariant:

$$\begin{aligned} T(x[n - n_0]) &= g[n]x[n - n_0] \\ &\neq y[n - n_0] = g[n - n_0]x[n - n_0] \end{aligned}$$

which is not TI.

- Memoryless: $y[n] = T(x[n])$ depends only on the n^{th} value of x , so it is memoryless.

(b) $T(x[n]) = \sum_{k=n_0}^n x[k]$

- Not Stable: $|x[n]| \leq M \rightarrow |T(x[n])| \leq \sum_{k=n_0}^n |x[k]| \leq |n - n_0|M$. As $n \rightarrow \infty$, $T \rightarrow \infty$, so not stable.
- Not Causal: $T(x[n])$ depends on the future values of $x[n]$ when $n < n_0$, so this is not causal.
- Linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= \sum_{k=n_0}^n ax_1[k] + bx_2[k] \\ &= a \sum_{k=n_0}^n x_1[k] + b \sum_{k=n_0}^n x_2[k] \\ &= aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

The system is linear.

- Not TI:

$$\begin{aligned} T(x[n - n_0]) &= \sum_{k=n_0}^n x[k - n_0] \\ &= \sum_{k=0}^{n-n_0} x[k] \\ &\neq y[n - n_0] = \sum_{k=n_0}^{n-n_0} x[k] \end{aligned}$$

The system is not TI.

- Not Memoryless: Values of $y[n]$ depend on past values for $n > n_0$, so this is not memoryless.

(c) $T(x[n]) = \sum_{k=n-n_0}^{n+n_0} x[k]$

- Stable: $|T(x[n])| \leq \sum_{k=n-n_0}^{n+n_0} |x[k]| \leq \sum_{k=n-n_0}^{n+n_0} x[k]M \leq (2n_0 + 1)M$ for $|x[n]| \leq M$, so it is stable.
- Not Causal: $T(x[n])$ depends on future values of $x[n]$, so it is not causal.
- Linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= \sum_{k=n-n_0}^{n+n_0} ax_1[k] + bx_2[k] \\ &= a \sum_{k=n-n_0}^{n+n_0} x_1[k] + b \sum_{k=n-n_0}^{n+n_0} x_2[k] = aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

This is linear.

- TI:

$$\begin{aligned} T(x[n - n_0]) &= \sum_{k=n-n_0}^{n+n_0} x[k - n_0] \\ &= \sum_{k=n-n_0}^n x[k] \\ &= y[n - n_0] \end{aligned}$$

This is TI.

- Not memoryless: The values of $y[n]$ depend on $2n_0$ other values of x , not memoryless.

(d) $T(x[n]) = x[n - n_0]$

- Stable: $|T(x[n])| = |x[n - n_0]| \leq M$ if $|x[n]| \leq M$, so stable.
- Causality: If $n_0 \geq 0$, this is causal, otherwise it is not causal.
- Linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= ax_1[n - n_0] + bx_2[n - n_0] \\ &= aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

This is linear.

- TI: $T(x[n - n_d]) = x[n - n_0 - n_d] = y[n - n_d]$. This is TI.
- Not memoryless: Unless $n_0 = 0$, this is not memoryless.

(e) $T(x[n]) = e^{x[n]}$

- Stable: $|x[n]| \leq M$, $|T(x[n])| = |e^{x[n]}| \leq e^{|x[n]|} \leq e^M$, this is stable.
- Causal: It doesn't use future values of $x[n]$, so it is causal.
- Not linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= e^{ax_1[n] + bx_2[n]} \\ &= e^{ax_1[n]} e^{bx_2[n]} \\ &\neq aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

This is not linear.

- TI: $T(x[n - n_0]) = e^{x[n - n_0]} = y[n - n_0]$, so this is TI.
- Memoryless: $y[n]$ depends on the n^{th} value of x only, so it is memoryless.

(f) $T(x[n]) = ax[n] + b$

- Stable: $|T(x[n])| = |ax[n] + b| \leq a|M| + |b|$, which is stable for finite a and b .
- Causal: This doesn't use future values of $x[n]$, so it is causal.
- Not linear:

$$\begin{aligned} T(cx_1[n] + dx_2[n]) &= acx_1[n] + adx_2[n] + b \\ &\neq cT(x_1[n]) + dT(x_2[n]) \end{aligned}$$

This is not linear.

- TI: $T(x[n - n_0]) = ax[n - n_0] + b = y[n - n_0]$. It is TI.
- Memoryless: $y[n]$ depends on the n^{th} value of $x[n]$ only, so it is memoryless.

(g) $T(x[n]) = x[-n]$

- Stable: $|T(x[n])| \leq |x[-n]| \leq M$, so it is stable.
- Not causal: For $n < 0$, it depends on the future value of $x[n]$, so it is not causal.
- Linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= ax_1[-n] + bx_2[-n] \\ &= aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

This is linear.

- Not TI:

$$\begin{aligned} T(x[n - n_0]) &= x[-n - n_0] \\ &\neq y[n - n_0] = x[-n + n_0] \end{aligned}$$

This is not TI.

- Not memoryless: For $n \neq 0$, it depends on a value of x other than the n^{th} value, so it is not memoryless.

(h) $T(x[n]) = x[n] + u[n + 1]$

- Stable: $|T(x[n])| \leq M + 3$ for $n \geq -1$ and $|T(x[n])| \leq M$ for $n < -1$, so it is stable.
- Causal: Since it doesn't use future values of $x[n]$, it is causal.
- Not linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= ax_1[n] + bx_2[n] + 3u[n + 1] \\ &\neq aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

This is not linear.

- Not TI:

$$\begin{aligned} T(x[n - n_0]) &= x[n - n_0] + 3u[n + 1] \\ &= y[n - n_0] \\ &= x[n - n_0] + 3u[n - n_0 + 1] \end{aligned}$$

This is not TI.

- Memoryless: $y[n]$ depends on the n^{th} value of x only, so this is memoryless.

2.26

The response of the system to delayed step:

$$\begin{aligned} y[n] &= x[n] * h[n] \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n - k] \\ &= \sum_{k=-\infty}^{\infty} u[k - 4]h[n - k] \\ y[n] &= \sum_{k=4}^{\infty} h[n - k] \end{aligned}$$

Evaluating the above summation:

For $n < 4$: $y[n] = 0$

For $n = 4$: $y[n] = h[0] = 1$

For $n = 5$: $y[n] = h[1] + h[0] = 2$

For $n = 6$: $y[n] = h[2] + h[1] + h[0] = 3$

For $n = 7$: $y[n] = h[3] + h[2] + h[1] + h[0] = 4$

For $n = 8$: $y[n] = h[4] + h[3] + h[2] + h[1] + h[0] = 2$

For $n \geq 9$: $y[n] = h[5] + h[4] + h[3] + h[2] + h[1] + h[0] = 0$

2.35. We first re-write the system function $H(e^{j\omega})$:

$$\begin{aligned} H(e^{j\omega}) &= e^{j\pi/4} \cdot e^{-j\omega} \left(\frac{1 + e^{-j2\omega} + 4e^{-j4\omega}}{1 + \frac{1}{2}e^{-j2\omega}} \right) \\ &= e^{j\pi/4} G(e^{j\omega}) \end{aligned}$$

Let $y_1[n] = x[n] * g[n]$, then

$$\begin{aligned} x[n] &= \cos\left(\frac{\pi n}{2}\right) = \frac{e^{j\pi n/2} + e^{-j\pi n/2}}{2} \\ y_1[n] &= \frac{G(e^{j\pi/2})e^{j\pi n/2} + G(e^{-j\pi/2})e^{-j\pi n/2}}{2} \end{aligned}$$

Evaluating the frequency response at $\omega = \pm\pi/2$:

$$\begin{aligned} G(e^{j\frac{\pi}{2}}) &= e^{-j\frac{\pi}{2}} \left(\frac{1 + e^{-j\pi} + 4e^{-j2\pi}}{1 + \frac{1}{2}e^{-j\pi}} \right) = 8e^{-j\pi/2} \\ G(e^{-j\frac{\pi}{2}}) &= 8e^{j\pi/2} \end{aligned}$$

Therefore,

$$y_1[n] = (8e^{j(\pi n/2 - \pi/2)} + 8e^{j(-\pi n/2 + \pi/2)})/2 = 8 \cos\left(\frac{\pi}{2}n - \frac{\pi}{2}\right)$$

and

$$y[n] = e^{j\pi/4} y_1[n] = 8e^{j\pi/4} \cos\left(\frac{\pi}{2}n - \frac{\pi}{2}\right)$$

§2.9.4

Differentiation in Frequency

$$x[n] \xleftrightarrow{F} X(e^{j\omega})$$

prove $n x[n] \xleftrightarrow{F} j \frac{d X(e^{j\omega})}{d\omega}$

$$\text{pf : } F\{nx[n]\} = \sum_{n=-\infty}^{\infty} nx[n]e^{-j\omega n} = \frac{1}{-j} \sum_{n=-\infty}^{\infty} x[n] \frac{d(e^{-j\omega n})}{d\omega} = j \frac{d}{d\omega} \left\{ \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \right\} = j \frac{dX(e^{j\omega})}{d\omega}$$

§2.9.6

The Convolution Theorem

$$x[n] \xleftrightarrow{F} X(e^{j\omega}), \quad h[n] \xleftrightarrow{F} H(e^{j\omega})$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

$$Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$$

$$\begin{aligned} \text{pf : } Y(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} y[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x[k] h[n-k] e^{-j\omega n} \\ &= \sum_{m=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x[k] h[m] e^{-j\omega(m+k)} \quad (m = n - k) \\ &= \sum_{m=-\infty}^{\infty} h[m] e^{-j\omega m} \sum_{k=-\infty}^{\infty} x[k] e^{-j\omega k} = H(e^{j\omega}) X(e^{j\omega}) \end{aligned}$$

§2.9.7

$$y[n] = x[n] w[n]$$

prove $Y(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta$

Pf.:

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n] w[n] e^{-j\omega n} \\ &= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} w[n] \left(\int_{-\pi}^{\pi} X(e^{j\theta}) e^{j\theta n} d\theta \right) e^{-j\omega n} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_{n=-\infty}^{\infty} w[n] X(e^{j\theta}) e^{-j(\omega-\theta)n} \right) d\theta \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) \left(\sum_{n=-\infty}^{\infty} w[n] e^{-j(\omega-\theta)n} \right) d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta \end{aligned}$$

2.25(c) Matlab solution

Solution225.m

```
x = [ ones(1, 5) ];
h = [ zeros(1,2) ones(1, 6) zeros(1,3) ones(1, 6)];
y = conv(h, x);
n = 0:1:length(y)-1;
stem(n,y)
xlabel('Time index n'); ylabel('Amplitude');
```

